

Exact Matching Problem.

A construction firm is facing three liabilities of \$1,000, due at the end of each of the years 1, 2, and 3. There are three bonds available to match these liabilities, as follows:

Bond I	A bond due at the end of year 1 with a coupon rate of 1% per year, valued at an effective annual yield rate of 14%.
Bond II	A bond due at the end of year 2 with a coupon rate of 2% per year, valued at an effective annual yield rate of 15%.
Bond III	A zero-coupon bond due at the end of year 3 valued at an effective annual yield rate of 18%.

Calculate the total purchase price of the three bonds required to exactly match the liabilities.